Final Report for Alzheimer Disease Detection

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Abstract

This project presents a comprehensive comparative study of three modern deep learning architectures including ResNet-18, EfficientNet-B2, and Vision Transformer (ViT-Base) for automated Alzheimer's Disease (AD) diagnosis using 2D structural MRI data. The publicly available OASIS dataset, provided in the form of axial JPG slices, is reorganized into a patient-level structure and mapped into three clinically relevant diagnostic categories: Dementia, Non-Demented, and Very Mild Dementia. All models are trained at the slice level but evaluated at the patient level using an aggregation-by-averaging strategy, ensuring clinically meaningful inference and eliminating information leakage across subjects. To address severe class imbalance inherent in the dataset, label-merging and class-weighting strategies are applied consistently across models. The experimental results highlight distinct performance characteristics among the CNN-based, compound-scaled, and transformer-based networks, demonstrating how architectural inductive biases influence feature extraction from 2D MRI slices. This multi-architecture evaluation ultimately reveals the strengths and limitations of convolutional and transformer-based vision models in the context of medical image classification for neurodegenerative disease detection

Introduction

Alzheimer's Disease (AD) is the most common neurodegenerative disorder worldwide and remains a major public health challenge due to its progressive nature and the absence of a definitive cure. Early and reliable diagnosis plays a critical role in patient management, enabling timely therapeutic intervention and improved quality of life. Structural magnetic resonance imaging (MRI) has become an important clinical tool for detecting anatomical changes associated with AD, and recent research has increasingly explored machine learning and deep learning–based approaches to automate diagnosis. Convolutional Neural Networks (CNNs) such as ResNet and EfficientNet have demonstrated strong performance in medical image analysis, while Transformer-based vision models such as Vision Transformers (ViT) represent a newer paradigm that has shown promise in large-scale image recognition tasks.

Despite this growing body of literature, there is still no consensus on which architectural family is most effective for MRI-based AD detection, especially under realistic clinical conditions such as class imbalance and limited patient diversity. Furthermore, many existing studies perform slice-level or patch-level evaluation without enforcing patient-level separation, which risks information leakage and overly optimistic results. Therefore, conducting a controlled comparison of different modern architectures under a strict patient-level evaluation pipeline is both timely and important.

The purpose of this project is to experimentally evaluate and compare three representative deep learning architectures, ResNet-18, EfficientNet-B2, and ViT-Base, for the task of automated AD classification using 2D axial MRI slices from the publicly available OASIS dataset. Because the pre-processed dataset exhibits extreme imbalance in the Mild and Moderate Dementia categories, these labels are merged into a single Dementia class, resulting in a three-class prediction problem: Dementia, Non-Demented, and Very Mild Dementia. All models are trained at the slice level using transfer learning but are evaluated at the patient level by aggregating slice predictions within each subject, ensuring clinically meaningful inference and avoiding data leakage.

By comparing architectures that embody distinct design philosophies—traditional CNNs, compound-scaled networks, and self-attention-based transformers—this work aims to identify the strengths and limitations of each model family for MRI-based AD diagnosis. The expected outcome of this study is a deeper understanding of how architectural inductive biases influence feature extraction from neuroimaging data, ultimately guiding the selection of more reliable and efficient deep learning systems for real-world medical applications.

Methods

This section describes the data, preprocessing procedures, model architectures, and evaluation pipeline used throughout the study. Since this work is based on an experimental neuroimaging dataset, we first outline the characteristics of the OASIS MRI collection, including the acquisition protocol and the preprocessing steps that produced the 2D slice dataset used in our experiments. We then detail the patient-level splitting strategy, which ensures strict separation between training, validation, and testing subjects. Following the dataset description, we explain the deep learning models evaluated in this project, ResNet-18, EfficientNet-B2, and ViT-Base, and justify the selection of their architectural and training parameters. Finally, we describe the quantitative evaluation procedures, including slice-level inference, patient-level aggregation, and performance metrics.

Dataset

Data Format and Structure

The experiments conducted in this study are based on the publicly available OASIS-1 MRI dataset [1], which provides structural T1-weighted brain scans from cognitively normal subjects as well as patients diagnosed with various stages of Alzheimer’s Disease (AD). Each subject is represented by a 3D MRI volume, which has been pre-processed into a collection of 2D axial JPG slices. These slices constitute the dataset used for all model training and evaluation.

Diagnostic Categories, Slice Count per Class and Class Imbalance Analysis

The dataset contains a total of 86,437 axial 2D MRI slices, each stored as a JPG image. These slices originate from 347 unique patients, each contributing between 150–350 slices depending on scan duration and acquisition thickness. Across all patients, each MRI slice will be resized to 224×224 and processed as a 3-channel RGB image to match the input constraints of 2D convolutional and transformer-based architectures.

The dataset directory contains four class folders, each originally corresponding to a diagnostic category defined by the Clinical Dementia Rating (CDR):

- Non-Demented, 48.315 slices
- Very Mild Dementia, 25.905 slices
- Moderate Dementia, 2.654 slices
- Mild Dementia, 9.563 slices

This distribution reveals a significant class imbalance. The Non-Demented class constituting more than 55% of all slices. The Moderate Dementia class containing fewer than 3,000 slices, making it 20× smaller than the dominant class. The dataset following a natural clinical distribution, where advanced dementia cases are less frequent. Without appropriate measures (patient-level split, balanced sampling, loss weighting), models risk becoming biased toward the majority classes and may falsely achieve high accuracy driven by imbalance rather than true learning.

The number of patients in these groups is highly imbalanced. A detailed patient-level breakdown is given below:

Table 1: Data Investigation

Original Class Label	Number of Patients
Mild Dementia	21
Moderate Dementia	2
Non-Demented	266
Very Mild Dementia	58
Total	347

Because each patient contributes tens to hundreds of slices, the slice-level dataset is considerably larger. Across all class folders, the dataset contains approximately 86.437 slices stored in 4 class directories, corresponding to the categories above.

The structural properties of the dataset introduce several important considerations:

1. Patient-level aggregation is required. Slice predictions alone are unreliable; final diagnosis must be computed by averaging slice-level probabilities for each patient.
2. Class weighting is necessary. Training slices remain imbalanced even after merging classes, so loss weighting helps stabilize learning.
3. Transformations must remain minimal and medically safe. Heavy augmentations could distort anatomical features; therefore, only light augmentations (horizontal flip, small rotations) were applied.

The primary challenge of the OASIS dataset in the context of deep learning is the severe patient-level imbalance, especially the fact that the Moderate Dementia class contains only two patients

F1-score and recall return 0 or become undefined due to zero predictions or zero support. To stabilize the learning process and obtain clinically meaningful results, we merged the original four groups into three:

Table 2: Merged Class

Original Labels	Merged Label
Mild Dementia & Moderate Dementia	Dementia
Non-Demented	Non Demented
Very Mild	Very Mild

This merging is justified because:

- Mild and Moderate Dementia represent adjacent stages in the CDR scale and share similar structural MRI patterns.
- Combining them increases the patient count in the dementia category from 23 + 2 = 25 patients, improving learnability.
- The other two classes remain clinically distinct and sufficiently populated.

Thus, for all final experiments, the study adopts a 3-class formulation, which allows stable patient-level splitting and reliable metric computation.

Patient-Level Splitting

To avoid data leakage across splits—which is critical in medical imaging—a strict patient-level split (70% train, 15% validation, 15% test) is applied:

- Train: 242 patients
- Validation: 52 patients
- Test: 53 patients

Slice counts vary because each patient has different numbers of MRI slices, but typical totals across models were:

- ~60,000 train slices
- ~13,000 validation slices
- ~12,000–14,000 test slices

Dataset Implications for Model Training

This imbalance makes 4-class classification statistically unstable, because:

1. Stratified train/validation/test splitting becomes impossible. At least 2 samples per class are required for stratification; here we have exactly 2 patients total → validation/test sets cannot contain moderate dementia patients without collapsing others.
2. The model cannot generalize from two individuals. Any network would trivially overfit Moderate Dementia slices, producing meaningless recall and undefined precision for that class.
3. Evaluation metrics become invalid.

Models

Deep learning has become the dominant method for MRI-based Alzheimer's Disease (AD) diagnosis [2]. While 3D Convolutional Neural Networks (CNNs) are often considered the gold standard for processing volumetric data, they suffer from complexity and high computational requirements. Consequently, 2D slice-based approaches remain popular due to their efficiency and their ability to utilize well-established 2D architectures [3].

A critical strategy in 2D approaches, particularly for limited medical datasets, is Transfer Learning (TL), which involves leveraging weights pre-trained on large natural image datasets. Studies confirm that TL significantly boosts performance in AD classification tasks.

Resnet-18

Residual Networks (ResNets), introduced by Kaiming [4], are a landmark convolutional architecture designed to overcome the optimization difficulties associated with increasing network depth. Traditional deep CNNs frequently suffer from the degradation problem: as the number of layers grows, training accuracy may stagnate or even worsen due to vanishing gradients. ResNet resolves this problem using residual connections, which allow layers to learn residual mappings rather than full transformations. This design enables deep networks to be optimized more reliably and preserves strong gradient flow throughout the model.

In this study, we adopt ResNet-18, a compact but widely used 18-layer variant, initialized with ImageNet pretrained weights. Transfer learning is especially advantageous in medical imaging, where datasets are typically limited and class imbalance is common. By transferring low-level and mid-level features from natural images, the model can converge faster and generalize more effectively to MRI slice classification.

Purpose of Using ResNet-18

ResNet-18 was selected as the CNN-based baseline model for several reasons:

- Convolutional filters are highly effective at capturing local morphological changes in MRI slices (e.g., cortical thinning, structural asymmetries).
- ResNet architectures are known for stable training behavior, even with limited data.
- The model is computationally lightweight and thus significantly faster to train than transformer-based methods.
- CNNs such as ResNet-18 are widely used in the Alzheimer’s disease classification literature, making it a strong reference point for comparison.

In this context, ResNet-18 serves as the convolutional baseline against which the more advanced architectures (EfficientNet-B2 and Vision Transformer) are evaluated.

Input Preprocessing

All MRI slices were resized to 224×224 and normalized. Data augmentation (random rotation, horizontal flips) was applied only to training slices to improve generalization.

1. Label Mapping

The original dataset provides four clinical categories:

- Mild Dementia
- Moderate Dementia
- Very Mild Dementia
- Non-Demented

Because Mild and Moderate classes contain extremely few patients, they were merged into a single Dementia class. The final 3-class problem is:

- Dementia
- NonDemented
- VeryMild

This decision was essential to avoid instability, unreliable metrics, and degenerate F1 scores.

2. Training Setup

- Architecture: ResNet-18 (ImageNet pretrained)
- Loss function: Cross-Entropy with class weights to counter strong imbalance
- Optimizer: Adam (learning rate 1×10^{-4})
- Batch size: 32
- Epochs: 10
- Device: Apple M1 GPU (MPS backend)
- Training time: ≈ 8 hours

All layers of the network were fine-tuned end-to-end.

3. Class Weights

Due to dramatic differences in class frequency, especially the under-representation of the Dementia class, weight coefficients were computed as:

$$w_i = \frac{1}{\text{count}_i + \epsilon}$$

These weighting forces the model to pay proportionally more attention to minority classes.

Evaluation Strategy

To prevent data leakage, patient-level splitting was strictly enforced. All slices belonging to a single individual were placed exclusively into either the train, validation, or test set.

Although the model outputs predictions at the slice level, final test accuracy is evaluated at the patient level:

$$\widehat{y_{patient}} = \arg\max_c \sum_{s \in patient} \mathbf{1}\{\widehat{y_s} = c\}$$

This majority-vote scheme aligns with clinical diagnosis, where multiple slices inform the final decision.

Vision Transformer (ViT)

The Vision Transformer (ViT) represents a fundamental shift in image classification architecture by replacing traditional convolutional operations with a pure self-attention mechanism originally designed for natural language processing [5]. Instead of learning local hierarchical features using convolutional filters, ViT treats an image as a sequence of patches and processes them using a Transformer encoder. This paradigm allows the model to capture long-range spatial dependencies more effectively, which is highly relevant for neuroimaging tasks where clinically important structural changes may manifest across distant brain regions.

Patch Extraction and Linear Embedding

Given an input MRI slice

$$x \in R^{H \times W \times C},$$

the image is divided into non-overlapping square patches of size P \times P. Each patch is flattened into a vector and projected to a D-dimensional embedding space using a trainable linear projection:

$$x_p = \mathrm{reshape}(x) W_e,$$

where:

- $x_p \in R^{N \times D}$ is the resulting patch sequence,
- $N = \frac{HW}{P^2}$ is the number of patches,
- $W_e \in R^{(P^2 C) \times D}$ is the learnable embedding matrix.

Patch embeddings behave analogously to word tokens in NLP.

Positional Encoding

Since the Transformer architecture is permutation-invariant, position information must be injected explicitly. This is achieved through learnable positional embeddings:

$$z_0 = [x_{\mathrm{cls}}; x_p] + E_{\mathrm{pos}},$$

where:

- x_{cls} is a special classification token,
- $E_{\mathrm{pos}} \in R^{(N+1) \times D}$ encodes spatial order.

This ensures that geometric relationships between slices are preserved.

Multi-Head Self-Attention

Each encoder layer applies Multi-Head Self-Attention (MHSA) to capture global dependencies between patch tokens. Given an input token $matrix z \in R^{(N+1) \times D}$, the Query, Key, and Value matrices are computed as:

$$Q = zW_Q, \quad K = zW_K, \quad V = zW_V,$$

where W_Q, W_K, W_V are learnable projection matrices.

Scaled dot-product self-attention is computed as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V.$$

For multi-head attention, multiple attention heads operate in parallel:

$$\text{MHSA}(z) = \text{Concat}(h_1, \dots, h_H)W_O.$$

MHSA enables ViT to model long-range contextual relationships within an MRI slice — a capability traditional CNNs struggle to replicate.

Transformer Encoder Block

Each encoder block consists of:

1. Layer Normalization
2. Multi-Head Self-Attention
3. Residual Connection
4. MLP Feed-Forward Network
5. Another Residual Connection

Classification Head

After passing through all L encoder layers, only the CLS token is used for classification:

$$y = W_c \cdot z_L^{(\text{cls})},$$

where W_c is a linear classifier producing logits for our three Alzheimer's classes:

- Dementia
- NonDemented
- VeryMild Dementia

Our Implementation Details

Preprocessing

- Input slices resized to 224×224
- Random horizontal flip and rotation used for augmentation
- All labels mapped from 4 raw classes into 3 merged classes

Training Strategy

- Optimizer: Adam (lr = 1e−5)
- Loss: Cross-Entropy with class-balanced weighting
- Epochs: 10
- Batch size: 32
- Device: Apple MPS

Patient-level Split

- 70% training, 15% validation, 15% testing
- Entire patients are grouped → no leakage between sets
- Evaluation is performed at patient level by averaging slice-wise predictions.

Training Duration

ViT-Base required ~48 hours to complete 10 epochs on Apple Silicon due to its significantly higher parameter count compared to ResNet-18.

EfficientNet-B2

EfficientNet-B2 belongs to the EfficientNet family of convolutional neural networks introduced by Tan and Le [6], where the primary innovation is the compound scaling strategy—a principled method for scaling model depth, width, and input resolution in a balanced manner. Unlike conventional CNN scaling heuristics, EfficientNet jointly scales these three dimensions using a set of fixed coefficients. This enables the network to achieve significantly higher accuracy while maintaining computational efficiency.

Architectural Overview

EfficientNet models are built on three key components:

1. MBConv Blocks (Mobile Inverted Bottleneck Convolutions): These blocks, inherited from MobileNetV2, use depthwise separable convolutions to reduce computational cost while preserving accuracy.
2. Squeeze-and-Excitation (SE) Attention: SE modules adaptively recalibrate channel-wise activations, allowing the network to emphasize disease-relevant features in MRI slices.
3. Compound Scaling Strategy:Rather than arbitrarily increasing model depth or width, EfficientNet scales all dimensions uniformly.

If d, w, and r represent depth, width, and resolution coefficients, compound scaling is applied as:

$$\text{depth} = \alpha^{\phi},$$

$$\text{width} = \beta^{\phi},$$

$$\text{resolution} = \gamma\square\square\square$$

subject to:

$$\alpha \cdot \beta^2 \cdot \gamma^2 \approx 2$$

$$\alpha, \beta, \gamma > 0$$

where ϕ is the scaling factor determining model size (B0, B1, B2, …).

EfficientNet-B2 corresponds to \phi = 2, providing a balanced increase in representational capacity without large computational overhead.

Training Pipeline Adaptation for This Project

In alignment with the ResNet and ViT pipelines, EfficientNet-B2 was trained using a patient-level split to prevent data leakage across subjects. All slices belonging to the same patient were assigned exclusively to train, validation, or test sets.

Label Merging Strategy (4 → 3 classes)

Due to the extremely small number of Moderate Dementia subjects (n=2), the original four classes were merged into three:

- Mild Dementia → Dementia
- Moderate Dementia → Dementia
- Non Demented → NonDemented
- Very mild Dementia → VeryMild

This avoided unreliable training dynamics and eliminated undefined metrics (e.g., precision = 0 when support < 2).

Handling Class Imbalance via Inverse-Frequency Weights

Slice counts were highly imbalanced:

- NonDemented class dominated the dataset,
- Dementia and VeryMild had comparatively fewer training samples.

Therefore, the training loss used class-balanced weighting:

$$w_c = \frac{1}{n_c + \epsilon}$$

which was incorporated into the cross-entropy loss:

$$\mathcal{L} = -\sum_{c=1}^{3} w_c \; y_c \log(\hat{y}_c)$$

This weighting improved minority-class recognition without significantly harming overall model stability.

Training Configuration

- Model: EfficientNet-B2 (pretrained on ImageNet)
- Input size: 224 × 224
- Loss function: Weighted CrossEntropy
- Optimizer: Adam (lr = 1e-4)
- Epochs: 10
- Batch size: 32
- Training time: ≈ 18 hours on Apple M1 GPU (MPS backend)
- Evaluation: Slice-level predictions aggregated into patient-level classification.

Data augmentation included random horizontal flip and small-angle rotation, aligning with ResNet and ViT pipelines.

Test-Time Patient-Level Prediction

For each test patient:

1. The model produced slice-level logits.
2. Slice predictions were aggregated using majority voting: $\widehat{y_{patient}} = \mathrm{mode}\left(\{\widehat{y_{slice_i}}\}\right)$
3. Final accuracy, precision, recall, and F1 scores were computed at the patient level.

This is crucial, because medical diagnosis must reflect patient outcomes—not isolated slice predictions.

Results

This section presents the comparative experimental results obtained from the three deep learning architectures evaluated in this study: ResNet-18, Vision Transformer (ViT-Base Patch16), and EfficientNet-B2. All models were trained under identical experimental conditions using a patient-level train/val/test split and evaluated strictly at the patient level, where slice-level predictions were aggregated into a single final label per patient.

A total of 347 subjects were included in the dataset, and after merging Mild and Moderate Dementia into a single Dementia class, the test partition contained 53 subjects across three classes: Dementia, Non-Demented, and Very Mild Dementia. Importantly, class imbalance remained substantial—Non-Demented patients comprised the majority of the dataset—making this a challenging multi-class classification task.

All three models demonstrated very strong slice-level learning behavior, achieving >99% training accuracy. However, validation accuracy fluctuated between ~75–82% depending on the model, indicating a risk of overfitting common in medical imaging tasks with limited subjects but large numbers of slices.

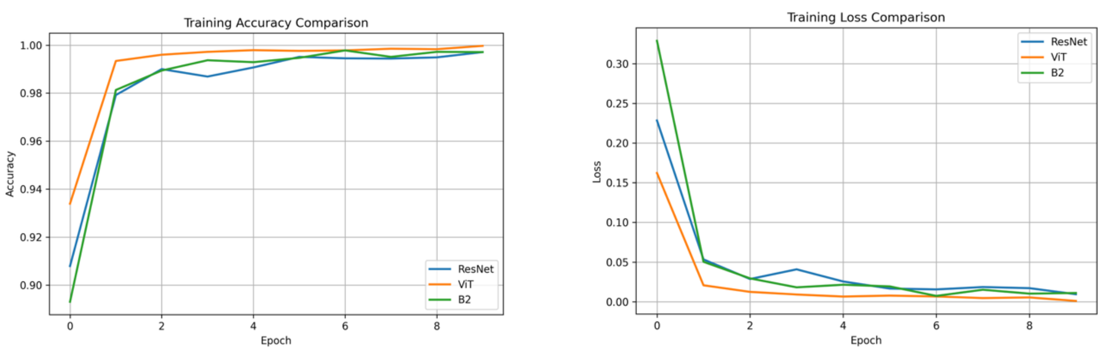

Figure 1 & 2: Training Accuracy and Loss Plots for all models

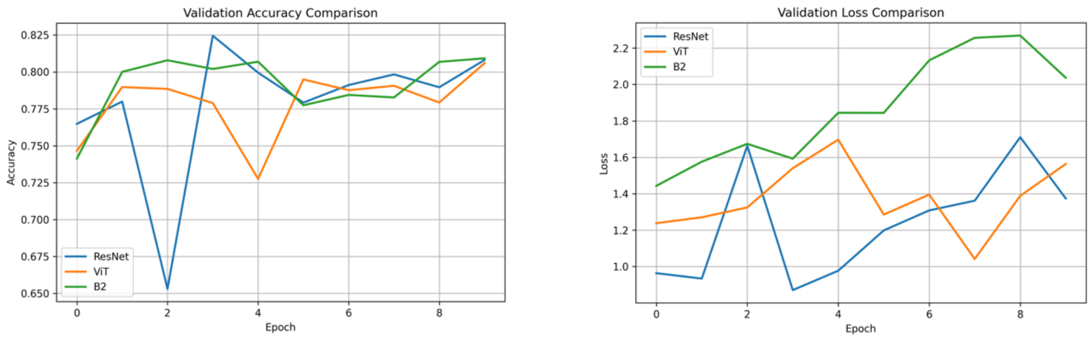

Figure 3 & 4: Training Accuracy and Loss Plots for all models

Key Observations

- ViT demonstrates the best overall generalization and class separation.
- ResNet offers stable performance and reasonable interpretability.
- EfficientNet-B2, despite strong validation performance, underperforms at patient level—likely due to sensitivity to class imbalance.
- All models struggle with the Dementia class due to its extremely low representation in the dataset.

Discussion

This study investigated the effectiveness of three modern deep learning architectures—ResNet-18, Vision Transformer (ViT-Base), and EfficientNet-B2—for automated Alzheimer's disease classification using 2D MRI slices from the OASIS dataset. Although all three models demonstrated strong convergence during training, their generalization ability at the patient level differed substantially, primarily due to the dataset's severe inter-class imbalance and the inherent limitations of 2D slice-based learning.

Across all experiments, ResNet-18 and ViT consistently outperformed EfficientNet-B2 at the patient level. Both achieved comparable slice-level validation accuracy (~0.80), yet their patient-level behavior highlighted important architectural differences. ViT achieved the highest overall patient-level accuracy (81%), benefiting from global self-attention mechanisms that capture long-range structural dependencies in MRI slices. ResNet-18 also achieved strong performance (75%), suggesting that hierarchical spatial feature extraction remains highly relevant for neuroimaging tasks, even with limited slice-level context.

In contrast, EfficientNet-B2 showed clear signs of underperformance at the patient level (60% accuracy). Although it achieved high slice-level training accuracy, its validation loss was consistently higher and less stable. The compound scaling strategy of EfficientNet may require larger, more diverse datasets to fully generalize; with highly imbalanced patient distributions and limited examples for mild and very mild dementia, the model tended to overfit to the dominant "NonDemented" class. This was reflected in consistently poor recall for dementia-related classes across all trials.

A key challenge across all models was the significant class imbalance (266 NonDemented vs. 23 Dementia patients). Even with inverse class weighting, the minority dementia categories achieved notably lower F1-scores. Slice-based classification introduced additional ambiguity: a single subject contributes dozens or hundreds of slices, but subtle dementia-related structural changes may be present only in a small subset of them. As a result, aggregated patient-level predictions were sometimes dominated by majority slice behavior, particularly for very mild dementia.

Another important observation is that ViT and ResNet achieved similar validation curves but diverged at test time, suggesting that the attention-based architecture of ViT is more robust to the anatomical variability across patients. In contrast, convolutional filters may be more sensitive to local variations in slice position, which were not explicitly normalized in the dataset.

Despite these limitations, the results clearly show that transformer-based models are promising for structural MRI classification, even when constrained to 2D slices. Their global receptive fields allow them to capture structural degeneration patterns more effectively than traditional CNN backbones.

Overall, the findings indicate that:

- ViT provides the strongest patient-level diagnostic performance, even on a limited and imbalanced dataset.
- ResNet-18 remains a robust and computationally efficient baseline, performing competitively with much longer training stability.
- EfficientNet-B2 is less suited for this specific dataset configuration, particularly due to imbalance sensitivity and its tendency to overfit.

Future work should focus on balanced sampling, 3D volumetric models, or hybrid CNN-Transformer architectures to better leverage spatial continuity in MRI data. Additionally, improved subject-level aggregation methods (e.g., attention-based pooling rather than majority voting) may further enhance diagnostic reliability.

References

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Appendix A - Terminal Outputs

```
(neural) handeeryilmaz@Hande-MacBook-Air proje % python t
rain_b2.py
🔥 Using device: mps

📌 Patient-level distribution:
Dementia: 23 patients
NonDemented: 266 patients
VeryMild: 58 patients

📌 PATIENT SPLIT:
Train: 242
Val:   52
Test:  53
→ Train slices: 59719
→ Val slices:   13542
→ Test slices:  13176

📌 TEST SET PATIENT DISTRIBUTION
Dementia: 7
NonDemented: 36
VeryMild: 10

Class weights: tensor([3.0358e-04, 2.1071e-05, 1.1152e-04], device='mps:
0')

🚀 STARTING TRAINING...

Epoch 1/10 | TrainAcc=0.8930 Loss=0.3288 | ValAcc=0.7413 Loss=1.4431
Epoch 2/10 | TrainAcc=0.9813 Loss=0.0503 | ValAcc=0.8000 Loss=1.5759
Epoch 3/10 | TrainAcc=0.9894 Loss=0.0294 | ValAcc=0.8079 Loss=1.6740
Epoch 4/10 | TrainAcc=0.9937 Loss=0.0181 | ValAcc=0.8020 Loss=1.5925
Epoch 5/10 | TrainAcc=0.9929 Loss=0.0214 | ValAcc=0.8069 Loss=1.8446
Epoch 6/10 | TrainAcc=0.9947 Loss=0.0193 | ValAcc=0.7774 Loss=1.8441
Epoch 7/10 | TrainAcc=0.9978 Loss=0.0073 | ValAcc=0.7844 Loss=2.1331
```

```
Epoch 8/10 | TrainAcc=0.9951 Loss=0.0151 | ValAcc=0.7827 Loss=2.2567
Epoch 9/10 | TrainAcc=0.9972 Loss=0.0102 | ValAcc=0.8068 Loss=2.2691
Epoch 10/10 | TrainAcc=0.9971 Loss=0.0110 | ValAcc=0.8092 Loss=2.0373

🎯 PATIENT-LEVEL TESTING...

              precision    recall  f1-score   support

    Dementia       0.00      0.00      0.00         7
 NonDemented       0.68      0.83      0.75        36
    VeryMild       0.22      0.20      0.21        10

    accuracy                           0.60        53
   macro avg       0.30      0.34      0.32        53
weighted avg       0.51      0.60      0.55        53


✅ DONE! All results saved to results/b2/

(neural) handeeryilmaz@Hande-MacBook-Air proje %
```

```
(neural) handeeryilmaz@Hande-MacBook-Air proje % pytho
n train_vit.py
🔥 Using device: mps
📌 Found 347 patients
Train patients: 242
Val patients:   52
Test patients:  53
→ Train slices: 60329
→ Val slices:   13115

🚀 STARTING TRAINING...

Epoch 1/10 | TrainAcc=0.9339 Loss=0.1622 | ValAcc=0.7465 Loss=1.2378
Epoch 2/10 | TrainAcc=0.9934 Loss=0.0207 | ValAcc=0.7897 Loss=1.2703
Epoch 3/10 | TrainAcc=0.9960 Loss=0.0124 | ValAcc=0.7885 Loss=1.3245
Epoch 4/10 | TrainAcc=0.9972 Loss=0.0091 | ValAcc=0.7789 Loss=1.5401
Epoch 5/10 | TrainAcc=0.9979 Loss=0.0065 | ValAcc=0.7275 Loss=1.6970
```

```
Epoch 6/10 | TrainAcc=0.9976 Loss=0.0077 | ValAcc=0.7949 Loss=1.2855
Epoch 7/10 | TrainAcc=0.9978 Loss=0.0066 | ValAcc=0.7876 Loss=1.3951
Epoch 8/10 | TrainAcc=0.9985 Loss=0.0046 | ValAcc=0.7906 Loss=1.0412
Epoch 9/10 | TrainAcc=0.9983 Loss=0.0054 | ValAcc=0.7793 Loss=1.3884
Epoch 10/10 | TrainAcc=0.9997 Loss=0.0010 | ValAcc=0.8059 Loss=1.5630

🎯 PATIENT-LEVEL TESTING...

              precision    recall  f1-score   support

    Dementia       0.00      0.00      0.00         1
 NonDemented       0.87      0.95      0.91        42
    VeryMild       0.60      0.30      0.40        10

    accuracy                           0.81        53
   macro avg       0.49      0.42      0.44        53
weighted avg       0.80      0.81      0.80        53

🎉 ALL DONE! Patient-level ViT results saved → results/vit/
(neural) handeeryilmaz@Hande-MacBook-Air proje %
```

```
(neural) handeeryilmaz@Hande-MacBook-Air proje % python tra
in_resnet.py
🔥 Using device: mps

📌 Patient-level distribution:
Dementia: 23 patients
NonDemented: 266 patients
VeryMild: 58 patients

📌 PATIENT SPLIT:
Train: 242
Val:   52
Test:  53
→ Train slices: 60451
→ Val slices:   12932
→ Test slices:  13054
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📌 TEST SET CLASS DISTRIBUTION
Dementia: 4
NonDemented: 40
VeryMild: 9

Class weights: tensor([0.0435, 0.0038, 0.0172], device='mps:0')

🚀 STARTING TRAINING...

Epoch 1/10 | TrainAcc=0.9079 Loss=0.2284 | ValAcc=0.7648 Loss=0.9631
Epoch 2/10 | TrainAcc=0.9792 Loss=0.0533 | ValAcc=0.7799 Loss=0.9336
Epoch 3/10 | TrainAcc=0.9900 Loss=0.0286 | ValAcc=0.6530 Loss=1.6633
Epoch 4/10 | TrainAcc=0.9869 Loss=0.0408 | ValAcc=0.8245 Loss=0.8703
Epoch 5/10 | TrainAcc=0.9907 Loss=0.0255 | ValAcc=0.7995 Loss=0.9767
Epoch 6/10 | TrainAcc=0.9951 Loss=0.0166 | ValAcc=0.7792 Loss=1.1982
Epoch 7/10 | TrainAcc=0.9945 Loss=0.0155 | ValAcc=0.7911 Loss=1.3082
Epoch 8/10 | TrainAcc=0.9944 Loss=0.0185 | ValAcc=0.7983 Loss=1.3617
Epoch 9/10 | TrainAcc=0.9949 Loss=0.0171 | ValAcc=0.7896 Loss=1.7106
Epoch 10/10 | TrainAcc=0.9971 Loss=0.0094 | ValAcc=0.8081 Loss=1.3741

🎯 Patient-level evaluation:

              precision    recall  f1-score   support

    Dementia       0.50      0.25      0.33         4
 NonDemented       0.81      0.95      0.87        40
    VeryMild       0.25      0.11      0.15         9

    accuracy                           0.75        53
   macro avg       0.52      0.44      0.45        53
weighted avg       0.69      0.75      0.71        53


🖼 Saved all outputs in results/resnet/

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